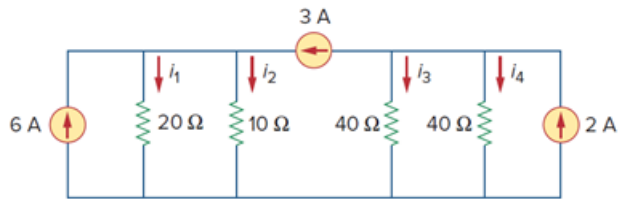


Problem

Given the circuit in Fig. 3.53, calculate the currents i_1 through i_4 .

Figure 3.53:



Step-by-step solution

Step 1 of 6

Draw the following circuit diagram shown in Figure 1.

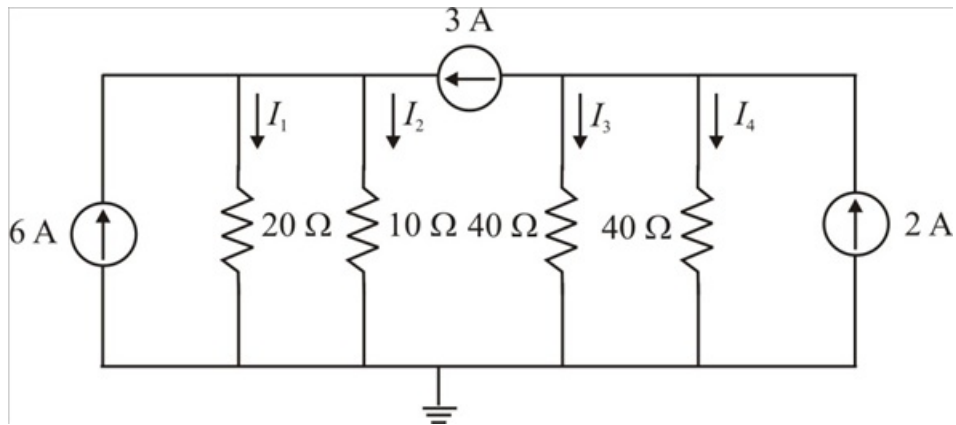


Figure 1

Step 2 of 6

The 20Ω and 10Ω resistors are in parallel. Their combined resistance is:

$$20\Omega \parallel 10\Omega = \frac{(20)(10)}{20+10} \\ = 6.66\Omega$$

The 40Ω and 40Ω resistors are in parallel. Their combined resistance is:

$$40\Omega \parallel 40\Omega = \frac{(40)(40)}{40+40} \\ = 20\Omega$$

Step 3 of 6

The reduced circuit with nodes is shown in Figure 2.

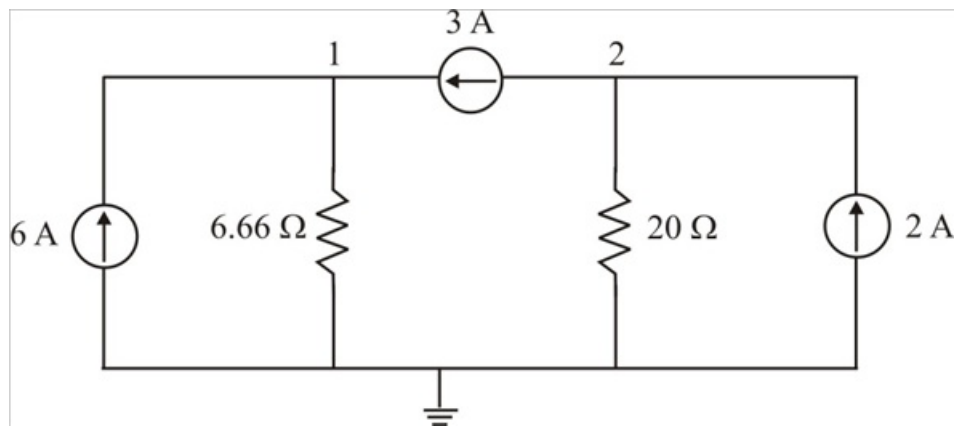


Figure 2

Step 4 of 6

Applying Kirchhoff's current law, the sum of currents entering the node is equal to the sum of the currents leaving from that node.

For the node 1, Kirchhoff's current law gives,

$$6 + 3 = i_a$$

$$i_a = 9 \text{ A}$$

This current flows through the 20 Ω and 10 Ω resistors in parallel.

Applying current division, the current i_1 can be calculated as:

$$\begin{aligned} i_1 &= \frac{10}{10 + 20}(i_a) \\ &= \frac{10}{10 + 20}(9) \\ &= 3 \text{ A} \end{aligned}$$

Therefore the current i_1 is .

Step 5 of 6

Applying current division, the current i_2 can be calculated as:

$$\begin{aligned} i_2 &= \frac{20}{10 + 20}(i_a) \\ &= \frac{20}{10 + 20}(9) \\ &= 6 \text{ A} \end{aligned}$$

Therefore the current i_2 is .

Step 6 of 6

Applying Kirchhoff's current law, the sum of currents entering the node is equal to the sum of the currents leaving from that node.

For the node 2, Kirchhoff's current law gives,

$$2 = 3 + i_b$$

$$i_b = -1 \text{ A}$$

This current flows through the 40 Ω and 40 Ω resistors in parallel.

Applying current division, the current i_3 can be calculated as:

$$\begin{aligned} i_3 &= \frac{40}{40+40}(i_b) \\ &= \frac{40}{40+40}(-1) \\ &= -0.5 \text{ A} \end{aligned}$$

Therefore the current i_3 is $\boxed{-0.5 \text{ A}}$.

Applying current division, the current i_4 can be calculated as:

$$\begin{aligned} i_4 &= \frac{40}{40+40}(i_b) \\ &= \frac{40}{40+40}(-1) \\ &= -0.5 \text{ A} \end{aligned}$$

Therefore the current i_4 is $\boxed{-0.5 \text{ A}}$.